

MORE NECTAR MEANS MORE ANTS AND MORE ANTS MEAN MORE NECTAR

MUTUALISMS

IN CONTEXT

KEY FIGURE

Dan Janzen (1939–)

BEFORE

1862 Charles Darwin proposes that an African orchid with a long nectar receptacle must be pollinated by a moth with an equally long proboscis.

1873 Belgian zoologist Pierre-Joseph van Beneden first uses the term “mutualism” in a biological context.

1964 The term “coevolution” is first used by American biologists Paul Ehrlich and Peter Raven to describe the mutualistic relations between butterflies and their food plants.

AFTER

2014 Researchers discover an unusual yet beneficial three-way mutualism involving sloths, algae, and moths.

In biology, there are several kinds of interaction between organisms. One species in an ecosystem may lose out to another when competing for the same resources. A prey species may be eaten by a predator. There are also symbiotic relationships, in which one species benefits but not at the expense of the other, or where one organism does not benefit but still survives. In